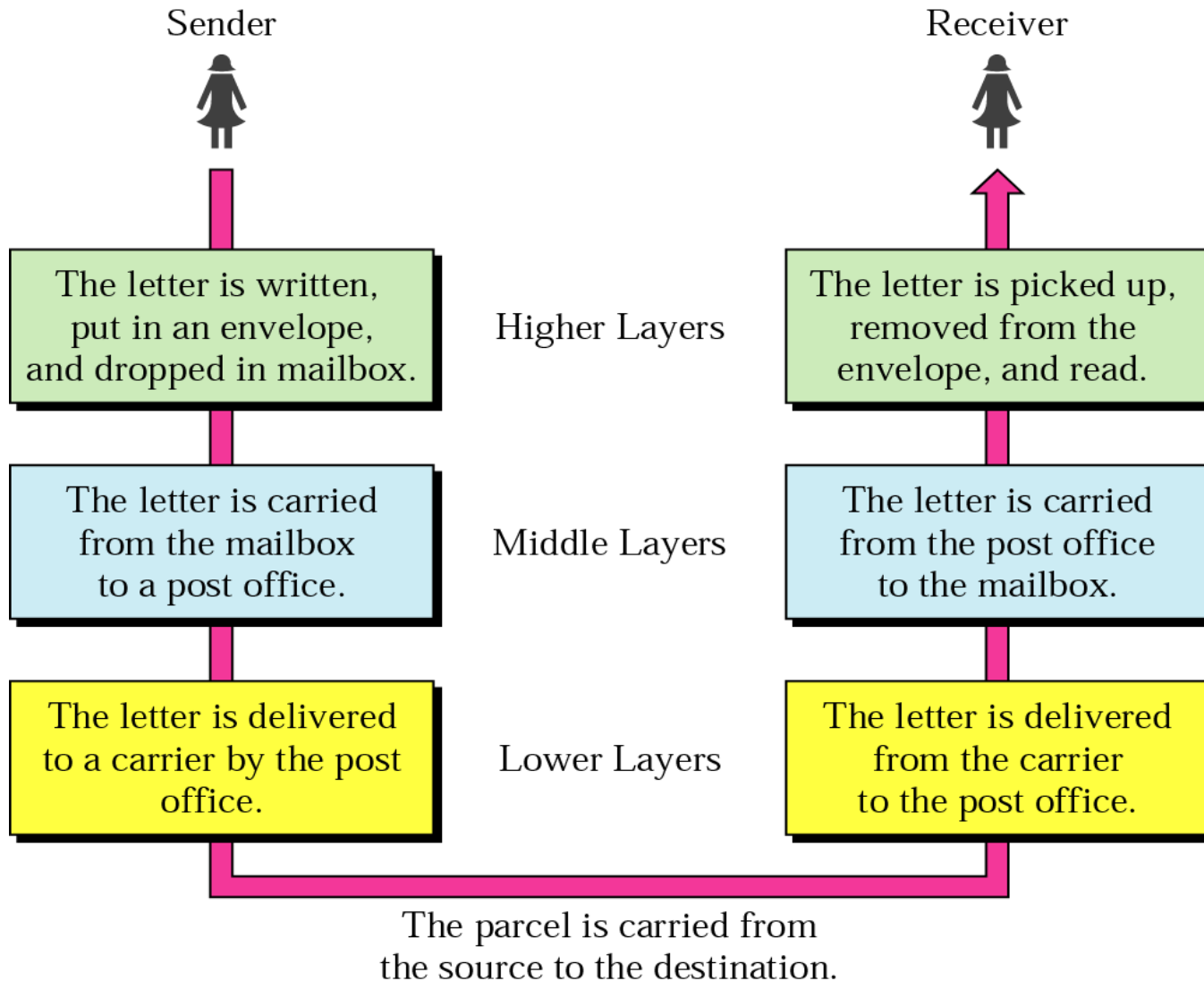


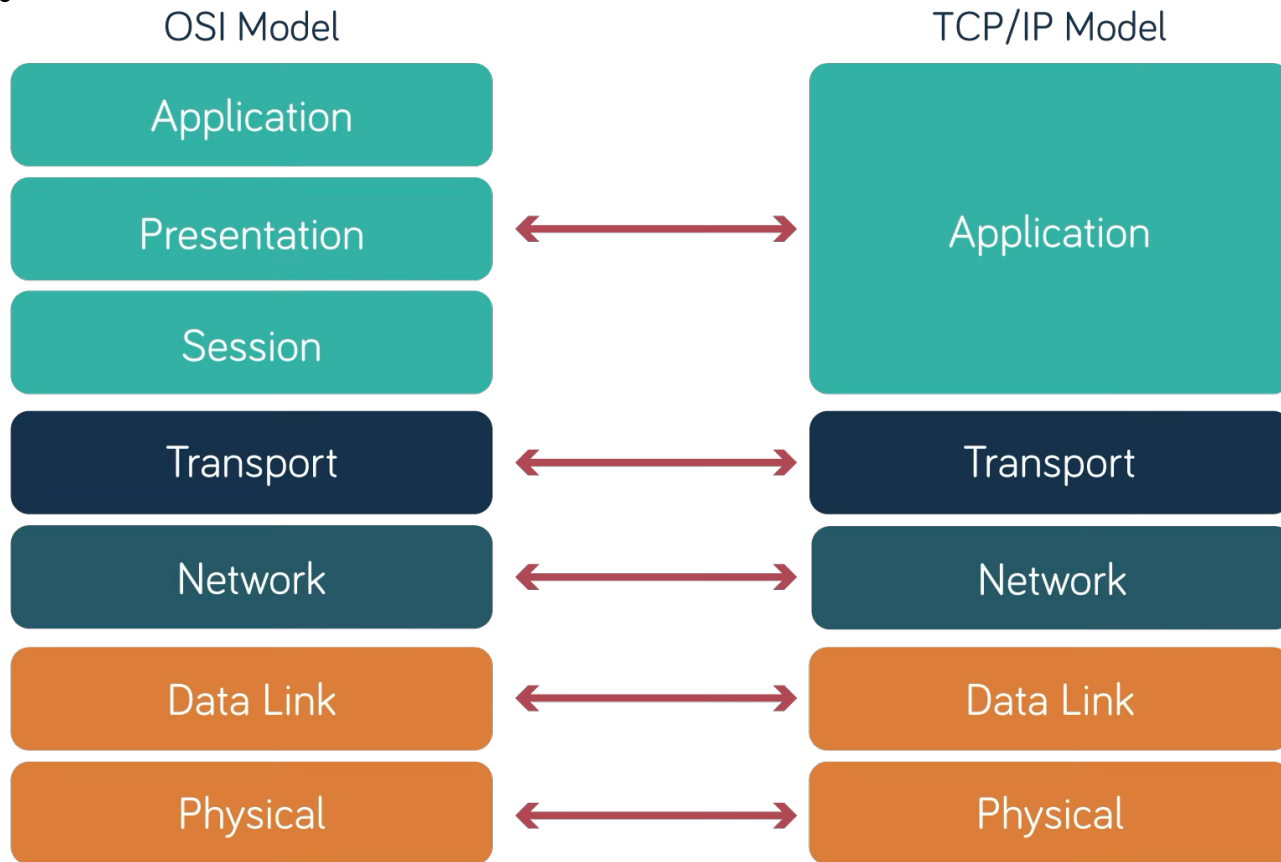
Network Layer Security *(IPSec)*

Basics of Layered Tasks: Sending a Letter



The OSI vs. the TCP/IP layers

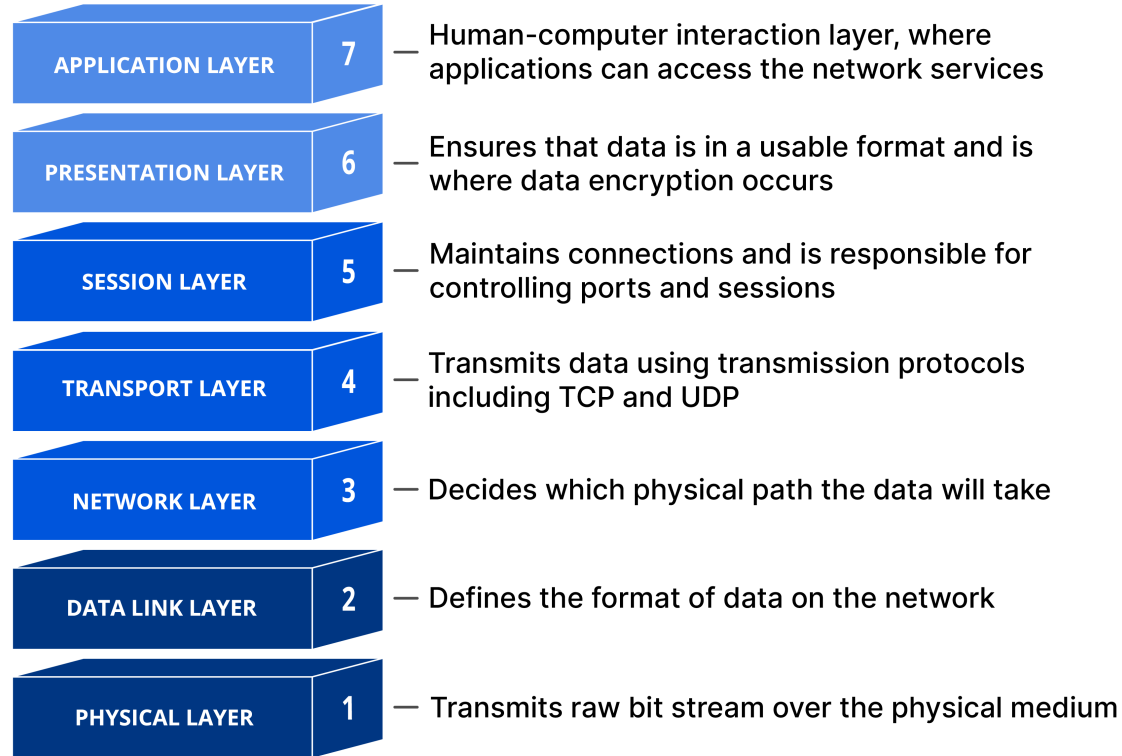
What are the 7 layers of the OSI Model in comparison with the 5 layers TCP/IP Model?



The application layer:

is the only layer that directly interacts with data from the user.

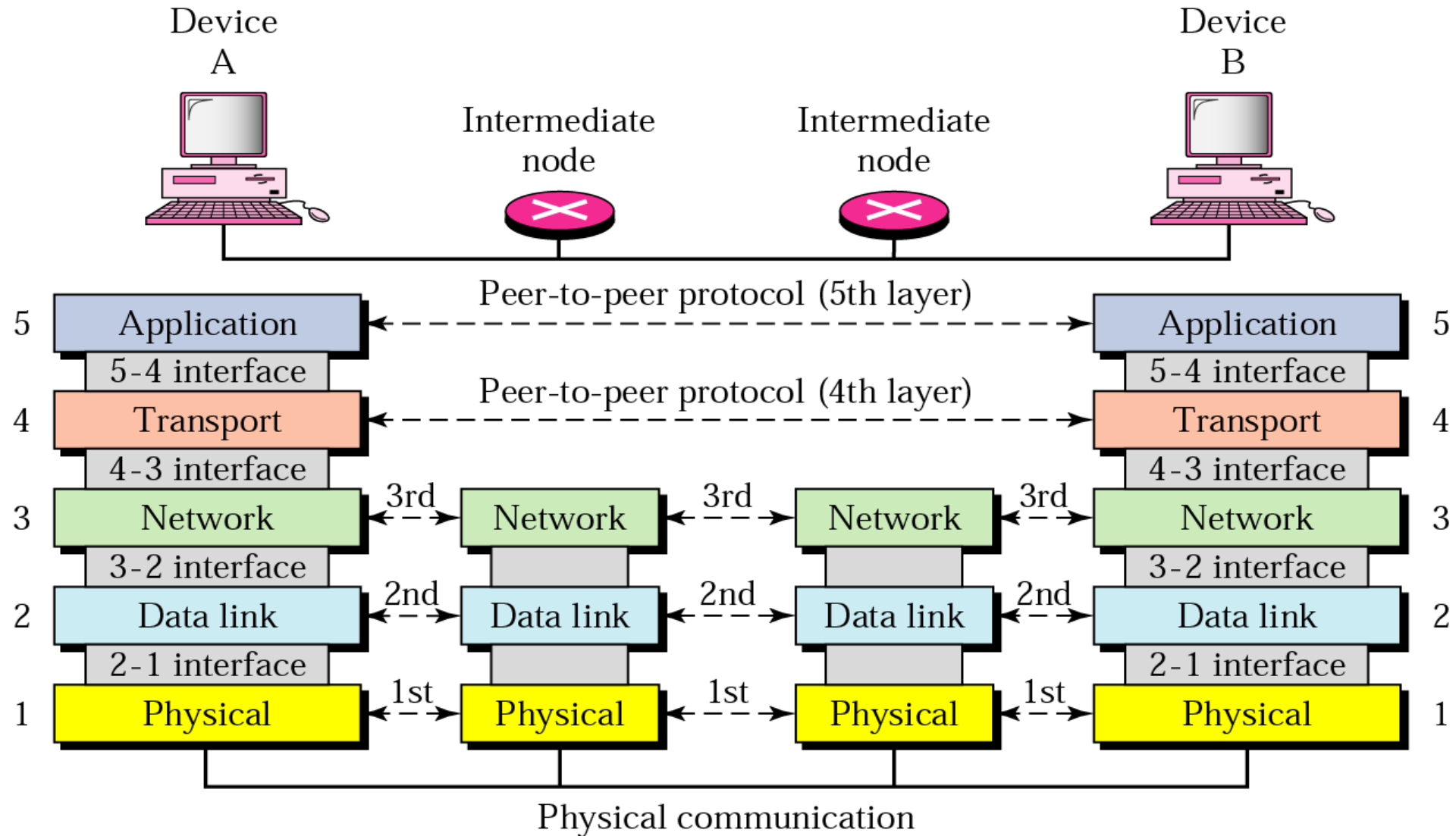
The Functions of the 7 Layers



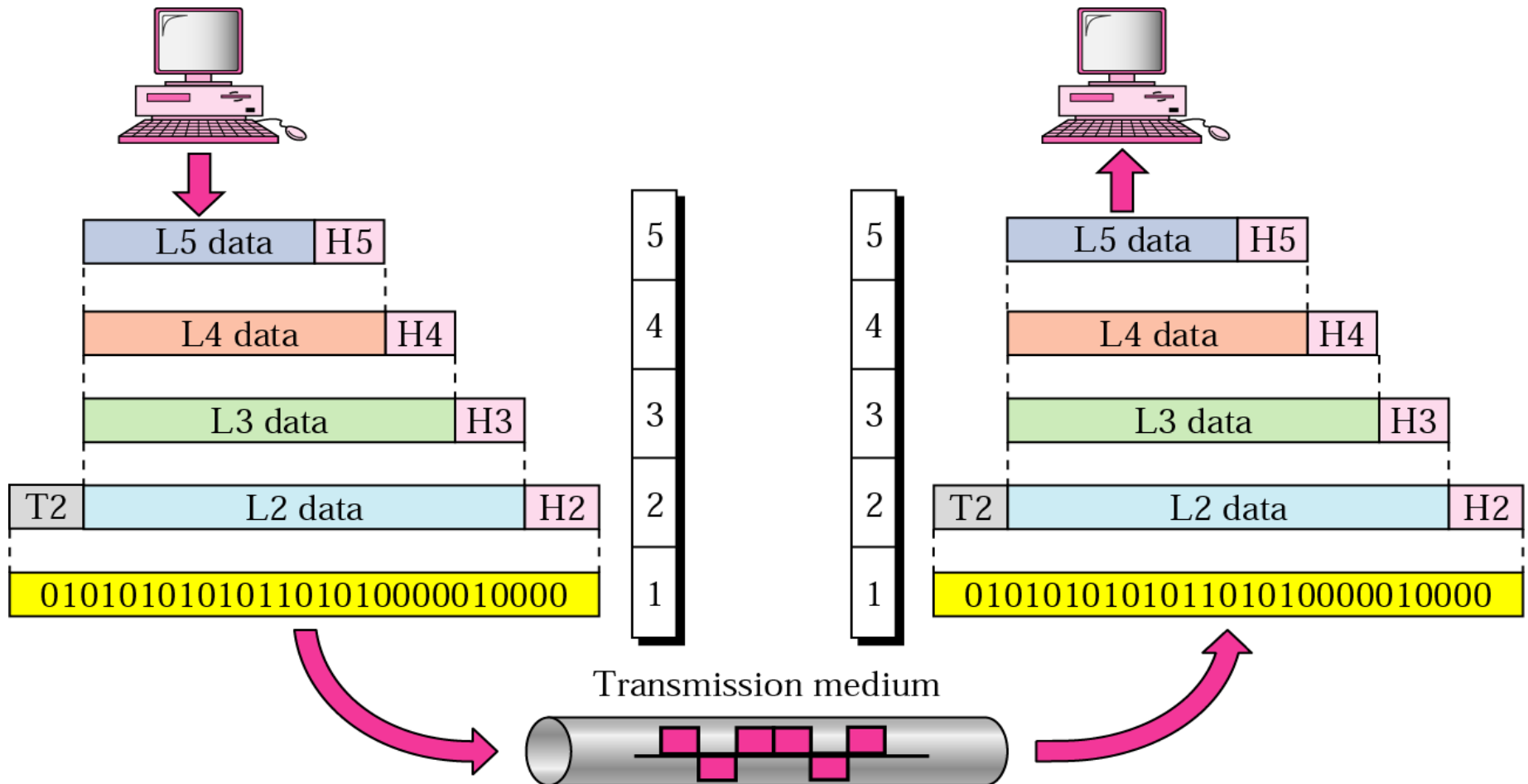
The Functions of the Internet Model

- The TCP/IP model is the foundation of the internet internal working,
- TCP ensures reliable data transmission through packet numbering, port numbers, and packet sequencing,
- IP enables the correct addressing and the routing of data packets from source to destination,
- The TCP/IP model has only five layers: physical, Data Link, network, transport, and application,
- It offers flexibility, platform independence, and high reliability for communicating heterogeneous devices

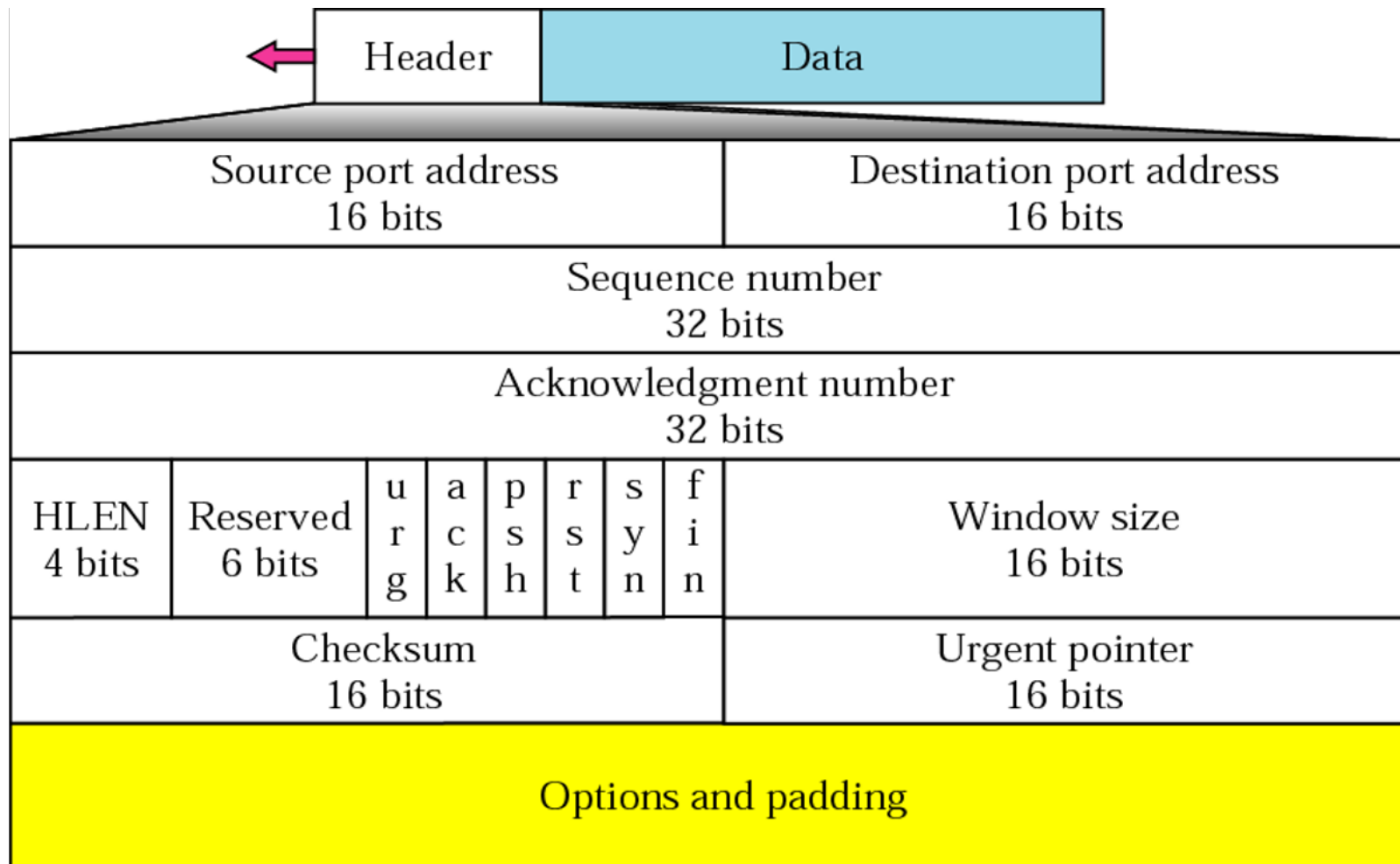
Peer-to-Peer Communication



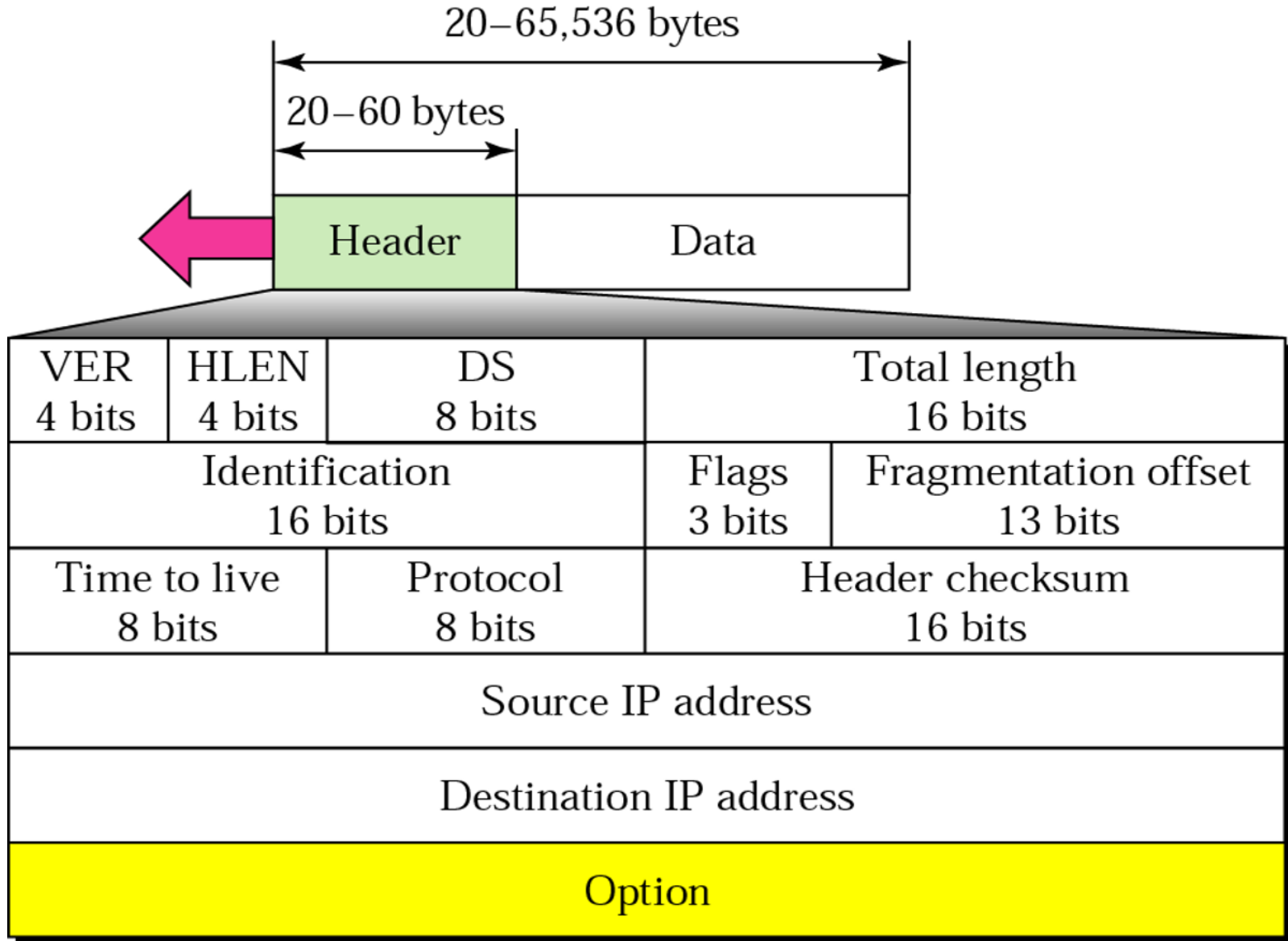
An Exchange using the Internet Model



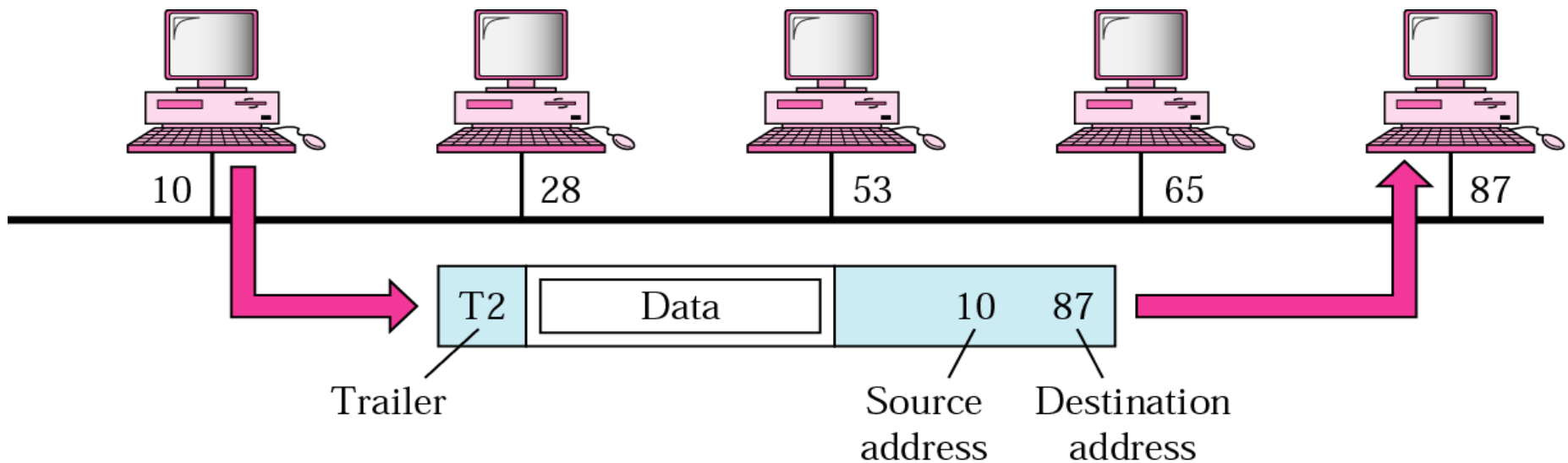
The TCP Segment Format



The IPv4 Datagram Format



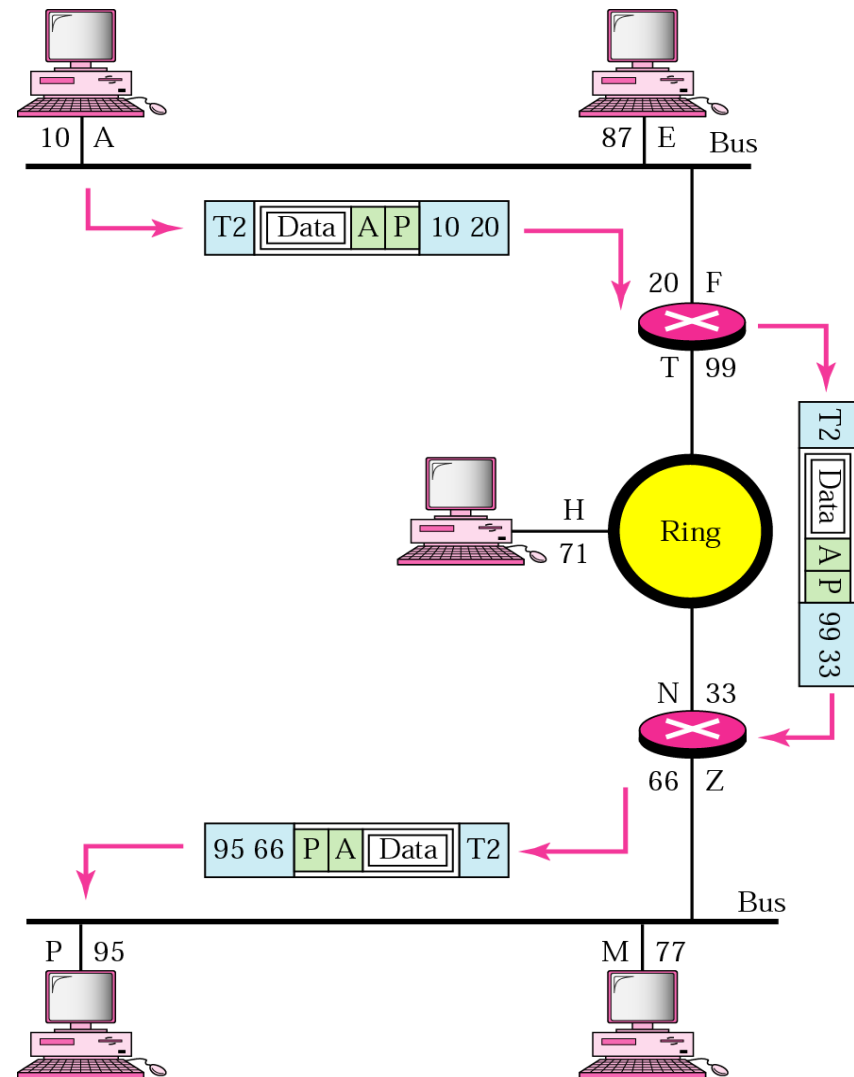
Example 1



In the above graph, a node with physical address 10 sends a frame to a node with physical address 87. The two nodes are connected by a link. At the data link level this frame contains physical addresses in the header. These are the only addresses needed. The rest of the header contains other information needed at this level. The trailer usually contains extra bits needed for error detection.

Example 2

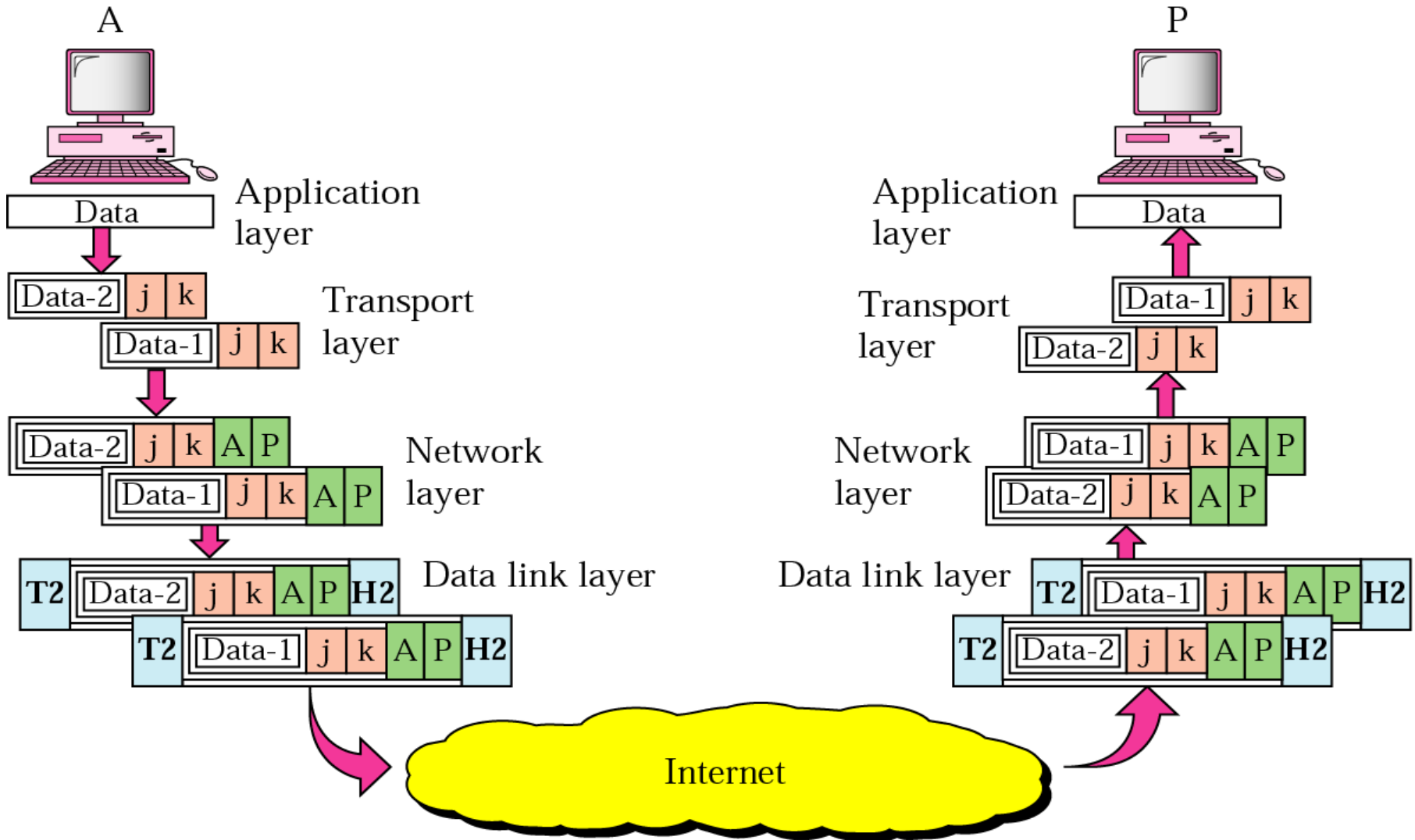
We want to send data from a node with network address A and physical address 10 to a node with a network address P and physical address 95. Because the two devices are located on different networks, we cannot use physical addresses only; the physical addresses only have local jurisdiction. What we need here are universal addresses that can pass through the LAN boundaries. The network address (logical) have this characteristic.



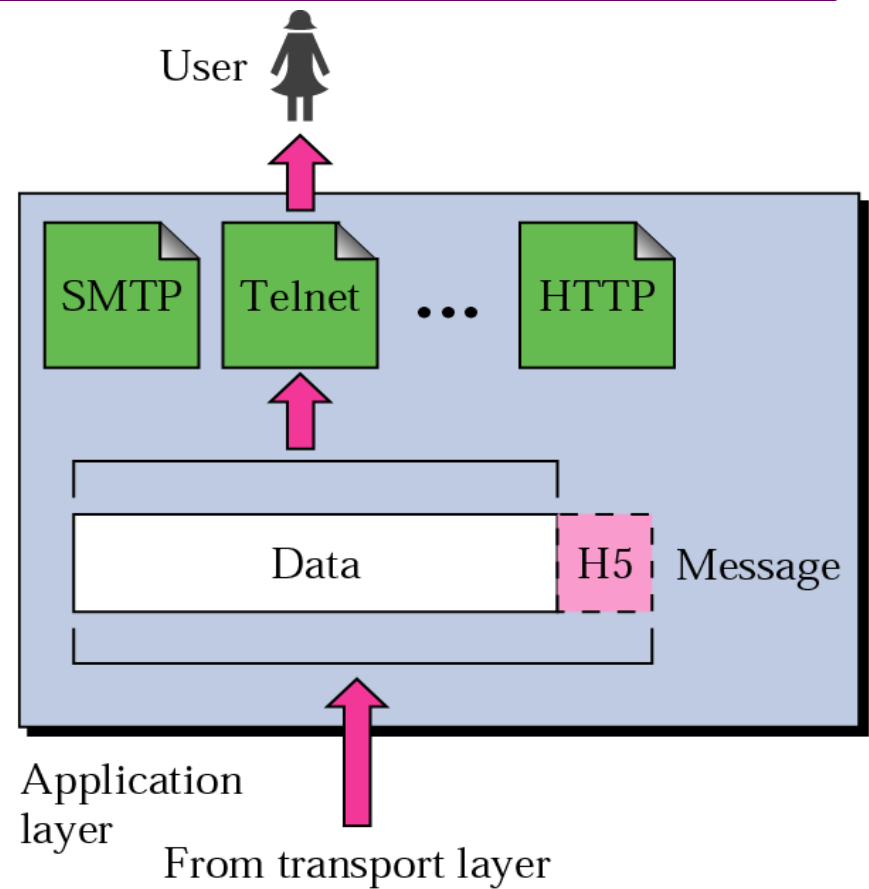
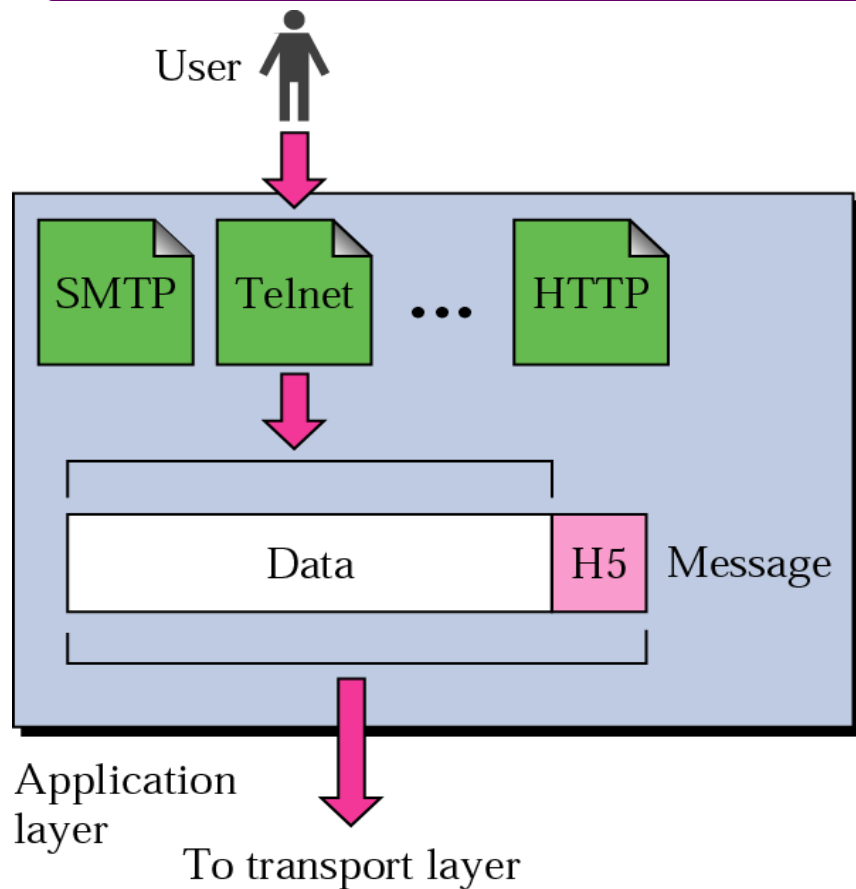
Example 3

The next slide shows an example of transport layer communication. Data coming from the upper layers have port addresses j and k (j is the address of the sending process, and k is the address of the receiving process). Since the data size is larger than the network layer can handle, the data are split into two packets, each packet retaining the port addresses (j and k). Then in the network layer, network addresses (A and P) are added to each packet.

Example 3 (Continue ...)



The Application Layer



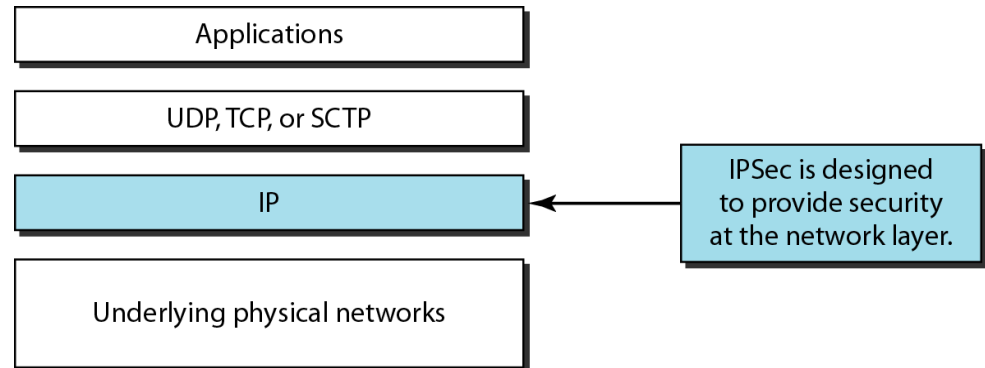
The application layer is responsible for providing services to the user.

IP Security

IP Security (IPSec) is a collection of protocols designed by the Internet Engineering Task Force (IETF) to provide security for a packet at the network level.

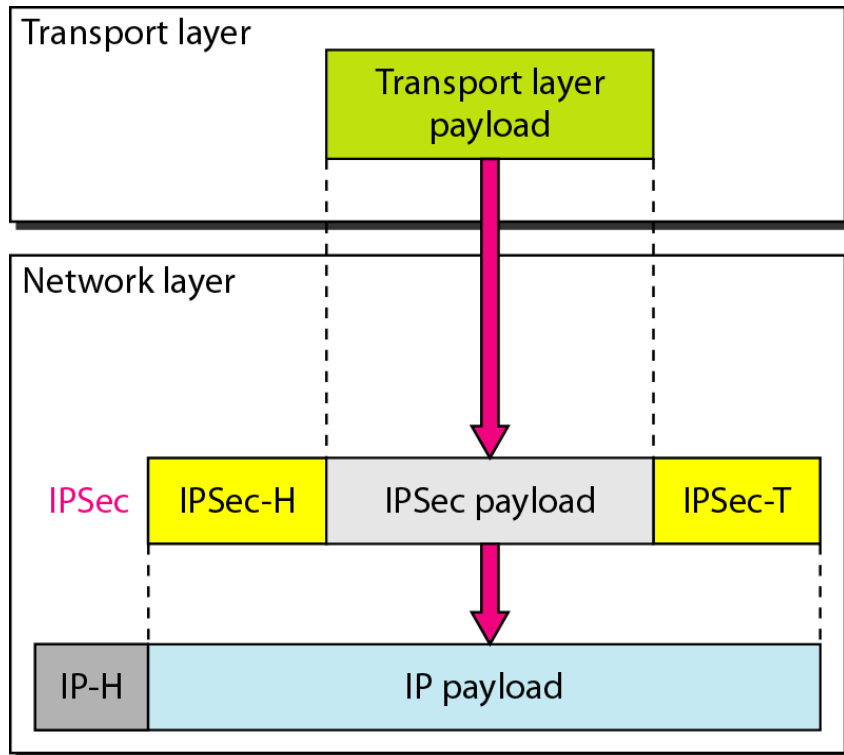
Topics in this section:

**Two Modes of IPSec
Two Security Protocols
Security Association
Virtual Private Networks**

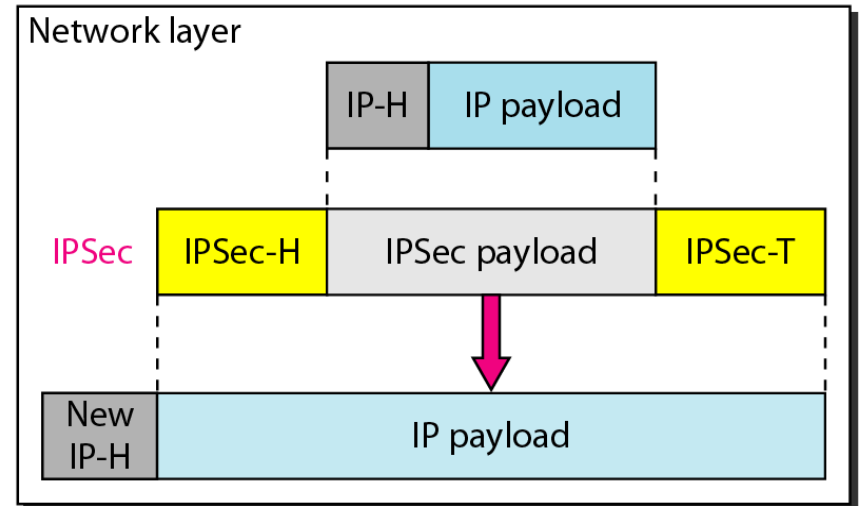


Transport Mode and Tunnel Mode of IPSec

IPSec in the transport mode does not protect the IP header; it only protects the information coming from the transport layer. IPSec in tunnel mode protects the original IP header



a. Transport mode



b. Tunnel mode

Virtual Private Networks (VPN)

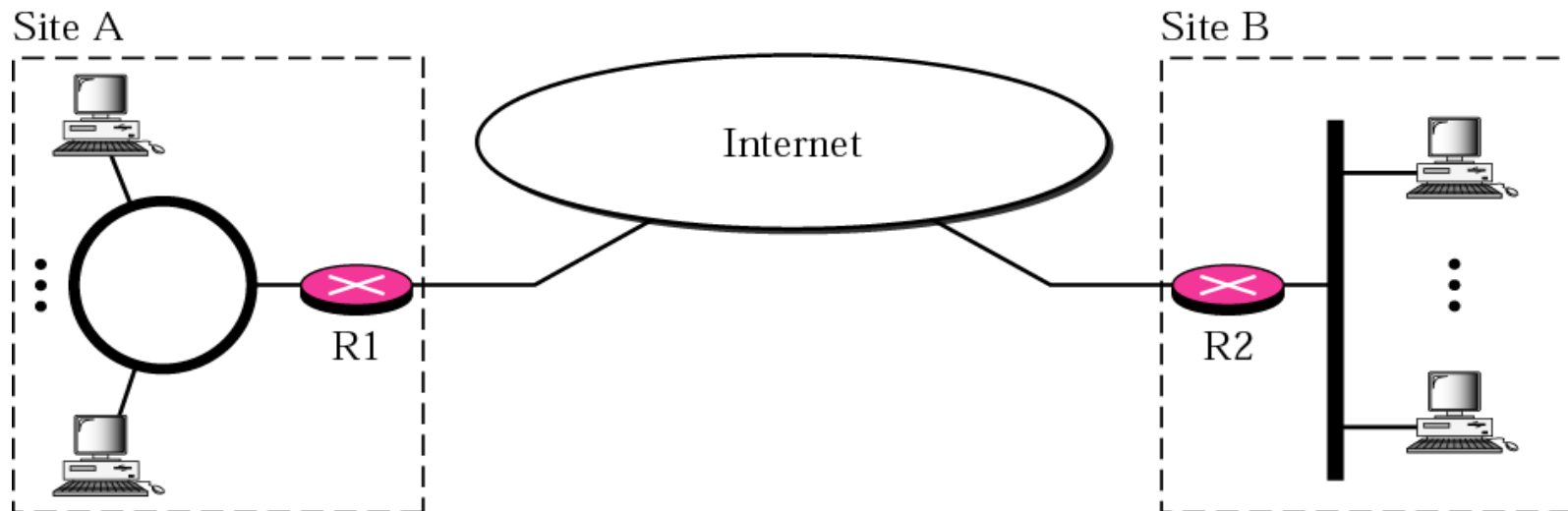
A private network is designed to be used only inside an organization. It allows access to shared resources and, at the same time, provides privacy.

Addresses for private networks

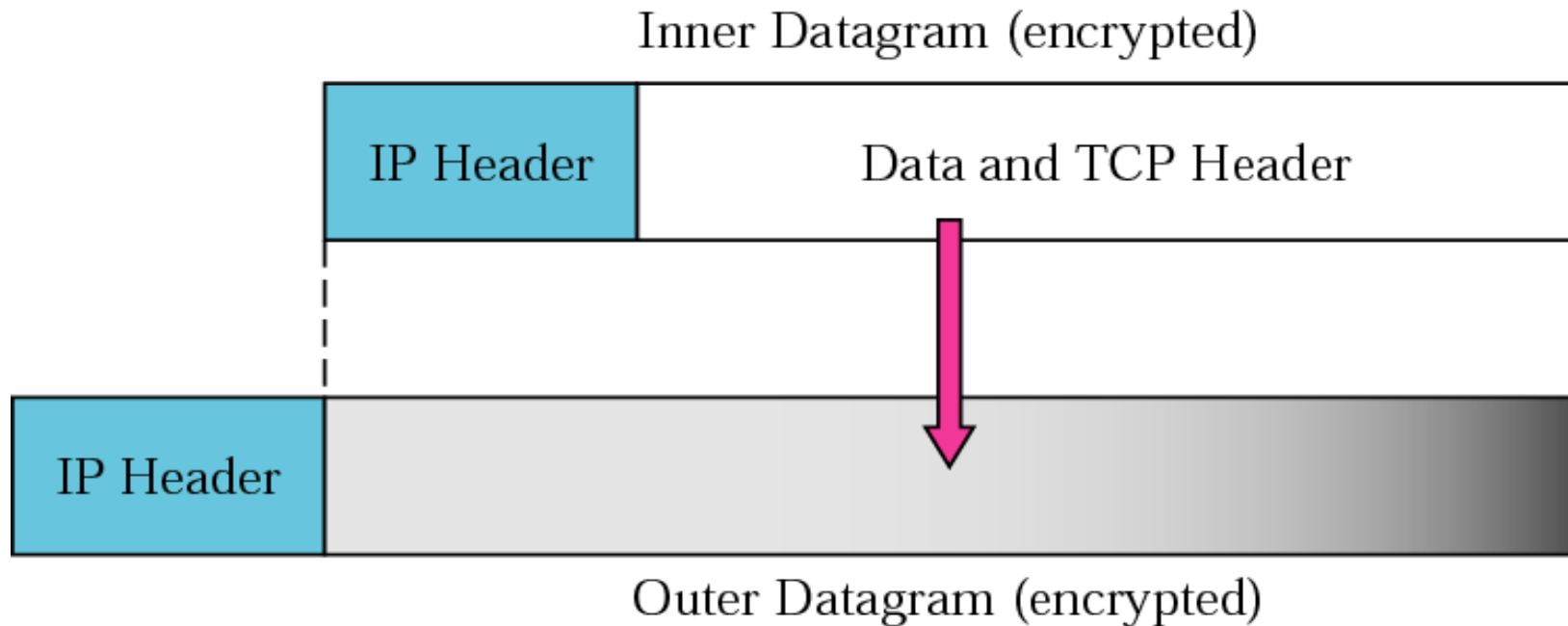
<i>Range</i>	<i>Total</i>
10.0.0.0 to 10.255.255.255	2^{24}
172.16.0.0 to 172.31.255.255	2^{20}
192.168.0.0 to 192.168.255.255	2^{16}

VPN is a technology for large organizations that use the global Internet for both intra- and inter-organization communication, but require privacy in their intra-organization communication.

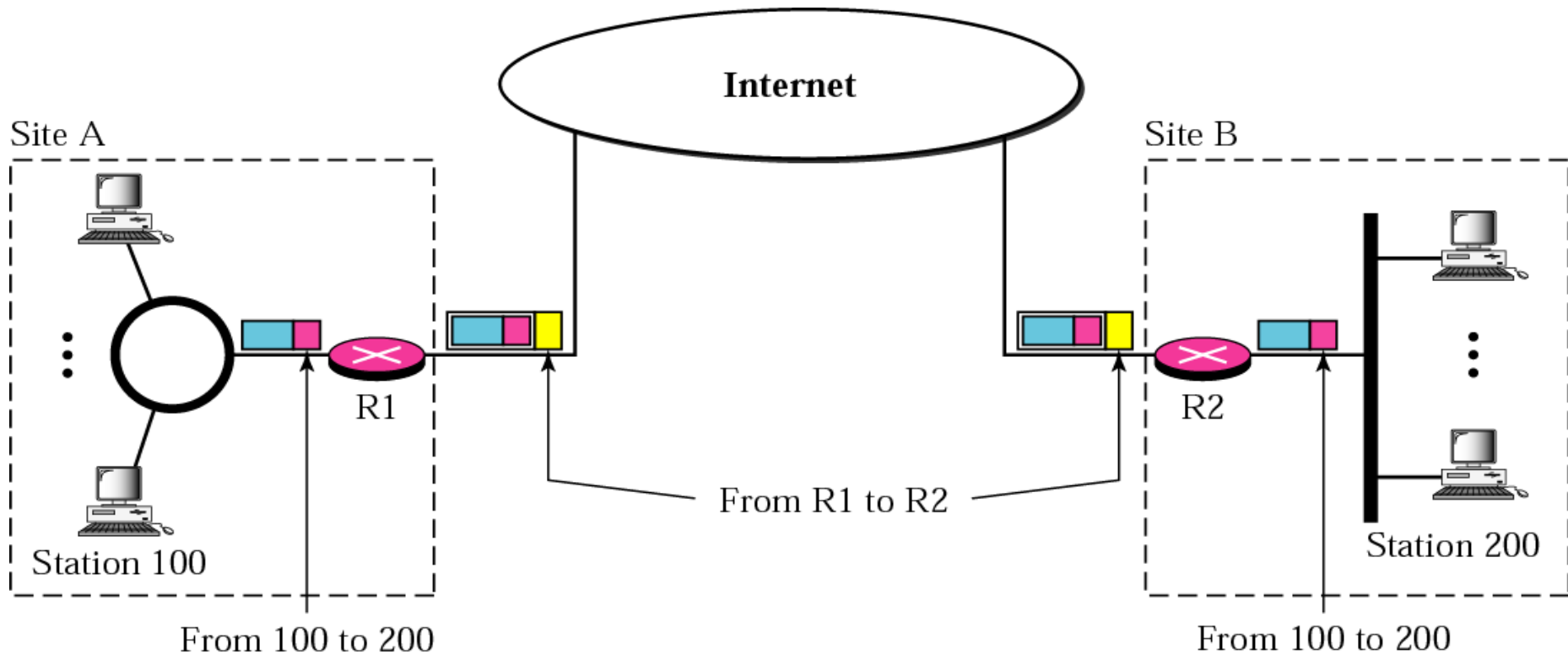
Virtual Private Networks (VPN)



Tunneling



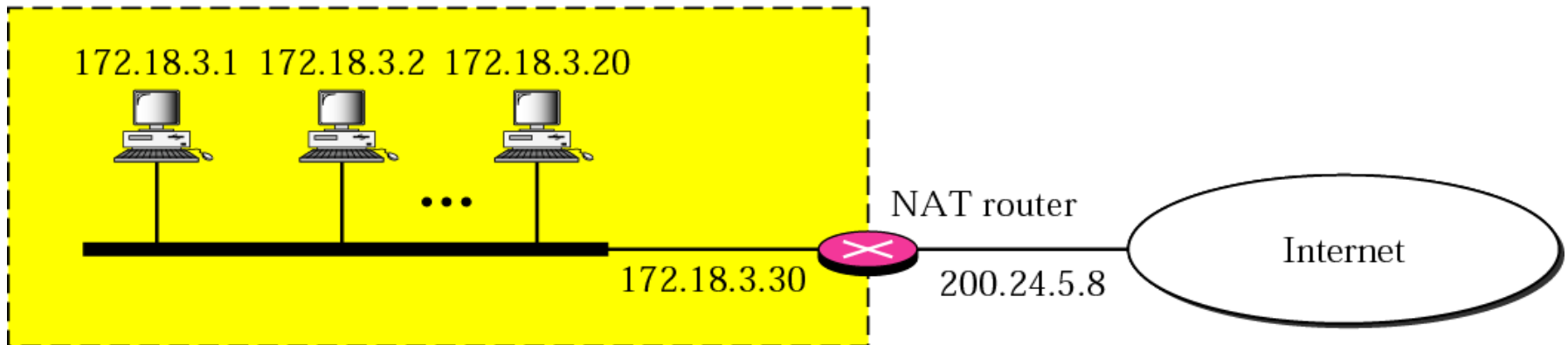
Addressing in a VPN



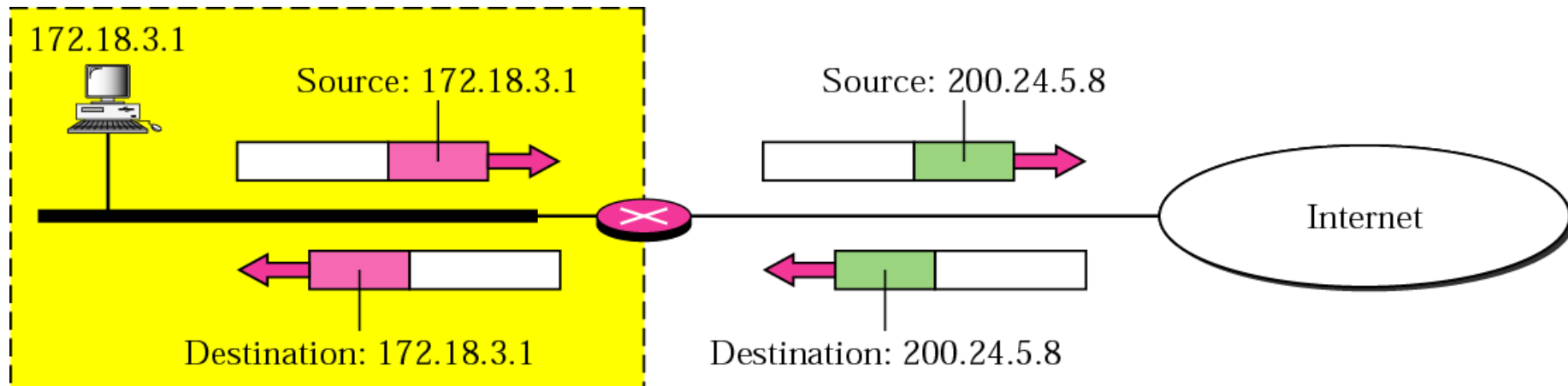
Network Address Translation (NAT)

Network address translation (NAT) allows a site to use a set of private addresses for internal communication and a set of global Internet addresses for communication with another site. The site must have only one single connection to the global Internet through a router that runs NAT software.

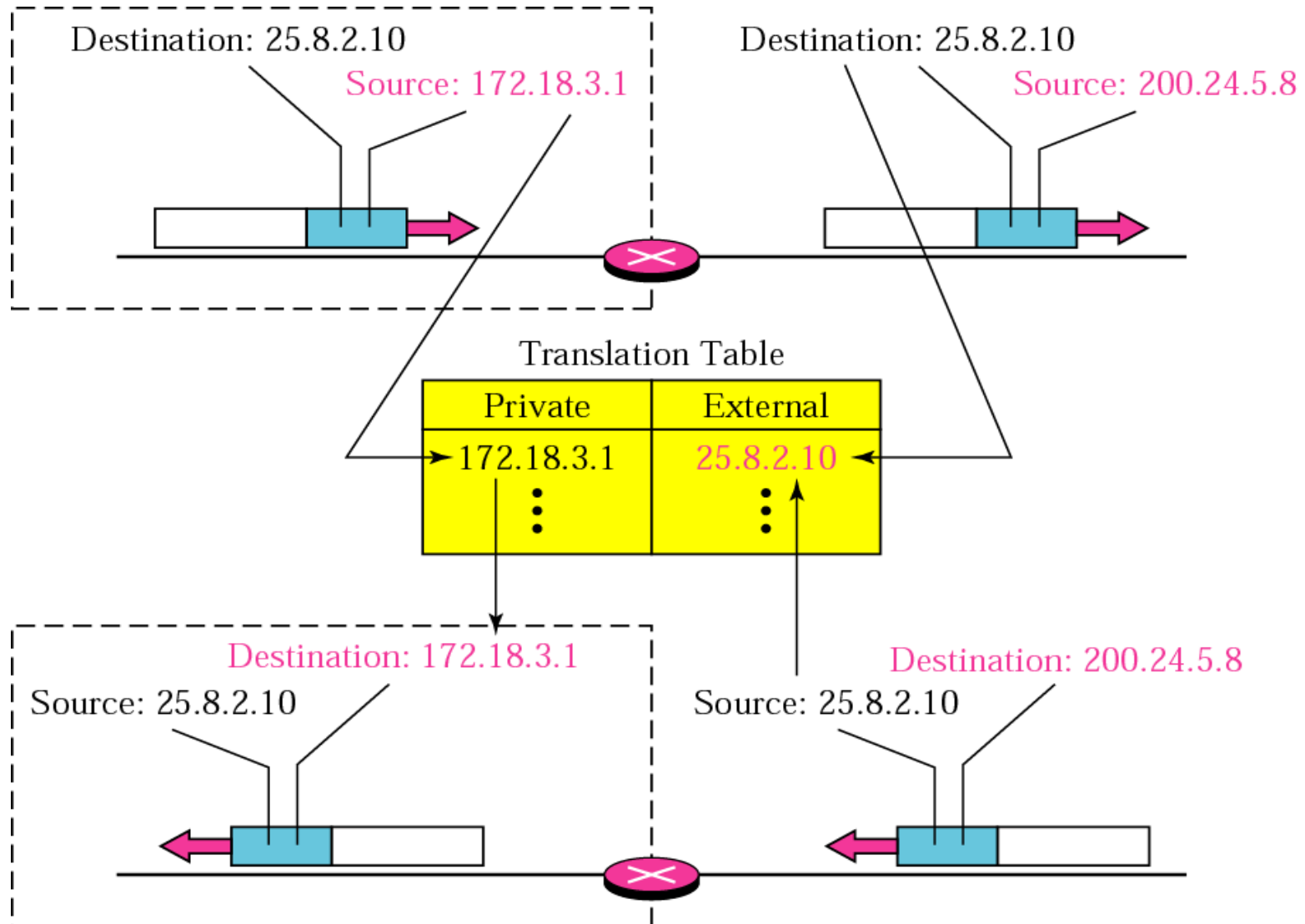
Site using private addresses



Network Address Translation (NAT)



Network Address Translation (NAT)



Network Address Translation (NAT)

<i>Private Address</i>	<i>Private Port</i>	<i>External Address</i>	<i>External Port</i>	<i>Transport Protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
...	...	...	...	...

